

VR-Sets. Preprint.

Part IX. Open Questions

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IX.0. Introduction

The concluding part of the preprint fixes the open questions of the VR-Sets programme. They are divided into three groups:

- (1) **Technical questions** — tasks of rigorously proving what has been formulated in the preprint as conjectures, and tasks of machine formalisation.
- (2) **Substantive extensions** — directions for developing the theory beyond the present preprint: connections with topoi, HoTT, extensions to MK and NBG, non-von-Neumann constructions.
- (3) **Philosophical questions** — substantive positions that have not received final formulation in the preprint.

Each question is formulated as a direction, not as a task with a known method of solution. Some of these questions may turn out to be technically routine (for instance, machine formalisation of already-proved theorems); others may require essentially new work.

IX.1. Technical questions

Question 1: Proof of Conjecture IV.1

Conjecture IV.1 asserts: the ZFC-mode of VR-Sets and classical ZFC are mutually interpretable (via a countable model of ZFC).

The direct direction was established in Theorem IV.1: the ZFC-mode is a model of ZFC. The converse direction — interpreting a countable model of ZFC in the ZFC-mode — requires:

- (a) the choice of a suitable countable model of ZFC (by Löwenheim–Skolem such models exist; a concrete construction convenient for operational interpretation is needed);
- (b) the construction of an operational representation of each element of this model as a describable functionality;
- (c) verification that the membership relation in the model is preserved under the passage to VR-Sets.

This is technical work feasible by standard means of model theory, but not trivial. An alternative approach — through the theorem on the existence of the minimal model of ZFC (L); the elements of L are describable by construction, which simplifies the operational interpretation.

Question 2: Proof of Conjecture IV.2

Conjecture IV.2 is the parallel of IV.1 for the ZFA-mode and the theory $ZFC^- + AFA$ (Aczel 1988). Technically similar, with additional difficulty on the AFA side: classical AFA theory also has an uncountable model, and a correspondence with the ZFA-mode requires work with a countable AFA-model.

Difficulty: in the AFA literature, countable models are less well studied than countable models of ZFC. A possible route — through Aczel's theorem on the final coalgebra of the power-set functor; the final coalgebra in a countable category may yield a countable AFA-model.

Question 3: Formalisation in Lean / Coq / Agda

Machine formalisation of the preprint in one of the proof assistants is the natural next task. This would provide:

- (a) a formal check of all proofs of Part III and Part IV;
- (b) a precise formulation of the definitions of Part II in the language of dependent types;
- (c) the possibility of machine verification of Conjectures IV.1 and IV.2 (at least the direct directions).

Lean 4 appears the most suitable due to its mathematical library (mathlib) and convenient support for coinductive definitions (for $\equiv$). Alternatively, Coq with its own library for non-well-founded sets, or Agda with its strong support for sized types.

Question 4: Precise characterisation of describable functionalities

The closure principle rests on the notion of an "operationally describable functionality." In the preprint this notion is used substantively but does not receive a precise formal definition. Possible formalisations:

- (a) through Turing machines: a describable functionality is one computable by a suitable machine;
- (b) through λ -calculus: a describable functionality is one expressible by a λ -term;
- (c) through primitive recursion and general recursion: a describable functionality is one given by a recursive definition.

These three formalisations are classically equivalent (by the Church–Turing thesis) but yield different practical approaches to working with VR-Sets. The choice of one or another formalisation is a separate technical question.

Question 5: Connection of $\equiv$ with bisimulation in rigorous formulation

In Part IV §IV.8 it was asserted that the operational identity $\equiv$ from Definition 4 structurally coincides with Aczel's bisimulation. This assertion requires a rigorous formulation: which sets/graphs are considered, in which category the correspondence operates, how this equivalence interacts with the ZFC and ZFA modes.

IX.2. Substantive extensions

Extension 1: Connection with topos theory

Categorical set theory (Lawvere's ETCS and its extensions) considers sets through the category $\mathbf{Set}$ with its universal properties. VR-Sets is closer to this tradition than to classical ZF: $\equiv$ operates as an isomorphism in the category of operational functionalities.

Open questions:

- (a) Is the VR-Sets universe (for the ZFC-mode) a topos? If so, what are its specific properties (for instance, does it contain a subobject classifier)?
- (b) What is the connection between the ZFC-mode and the category of sets in a suitable topos of computable functions?
- (c) Can the ZFA-mode be obtained as a topos with a suitable modification (without the well-foundedness condition on structural morphisms)?

These questions form a bridge to the powerful categorical apparatus; their development could significantly extend the expressive power of VR-Sets.

Extension 2: Homotopy type theory (HoTT)

In HoTT, equality is not a relation but a type (path). This is consistent with the operational ontology: $\equiv$ in VR-Sets is not a static relation but potentially a structure (two operationalities are equal in a particular way, and there may be many such ways).

Possible directions:

- (a) Reformulate VR-Sets in the language of HoTT, where $\equiv$ becomes a path type.
- (b) Investigate whether this yields new objects — operational sets with a non-trivial identity structure.
- (c) Connect the hierarchy of t-applications with n-types of HoTT.

This is a substantively close direction, but it would require considerable work in translating VR-Sets into a type calculus.

Extension 3: Morse–Kelley and NBG

In classical set theory, beyond ZFC, extensions with proper classes are considered: NBG (von Neumann–Bernays–Gödel) and Morse–Kelley. In these theories, "collections" too large to be sets are admitted.

In VR-Sets, the entire universe is countable, and so "too large collections" do not arise in the classical sense. However, operational analogues are possible:

- (a) **Describable families of functionalities** that are not sets through failure of operational describability. For instance, "all terminating algorithms computing reals" — this is not a set in VR-Sets, since the corresponding family is not enumerable.
- (b) **The hierarchy of meta-levels through t**: each $t^n(\omega)$ is a set, but "the entire hierarchy" is an unbounded process, not a separate object.

A substantive development of the operational analogue of class theory is an open direction.

Extension 4: Non-von-Neumann codings of numbers

In Part V, VR numbers are represented as von Neumann ordinals $O_n = \{O_0, \dots, O_{n-1}\}$. This is a standard coding but not the only one. Zermelo proposed an alternative: $\{\emptyset\}$, $\{\{\emptyset\}\}$, $\{\{\{\emptyset\}\}\}$, ...

In VR-Sets more exotic operational codings of numbers are possible — for instance, through infinite non-composite functionalities, through coinductive structures in the ZFA-mode, or through binary representations. Each such coding is a distinct operational realisation of $\mathbb{N}$; all are isomorphic as algebraic structures but differ in operational characteristics (operational depth, describability, and so on).

Open question: is there, among these codings, one distinguished in VR-Sets by a structural property (for instance, minimising operational depth or complexity of description)?

Consistency with §V.6. Part V §V.6 asserts that, under the standard von Neumann coding, VR numbers do not use the ZFA-mode. The present extension does not contradict this: it considers alternative non-von-Neumann codings as an open direction. Coinductive ZFA-mode codings of numbers, if developed, would be a separate operational realisation of $\mathbb{N}$ alongside the standard one — not a revision of the standard coding.

IX.3. Philosophical questions

Question 1: The nature of "action"

The slogan "only $\emptyset$ is, all else is doing" appeals to the notion of "action" as a fundamental ontological modality. In Part VIII this was indicated as a parallel to Whitehead's process ontology, but it was not developed in full.

Open questions:

- (a) What does "action" mean in the VR-Sets ontology, given that time does not enter it? Operational functionalities are integral objects (Part IV §IV.1), not temporal processes. Then "action" is what?
- (b) Can "action" be more ontologically fundamental than "object"? If so, this requires serious philosophical work not carried out in the preprint.
- (c) The relation to other ontologies of action (Heraclitus, Bergson, Whitehead, process theology) is an open philosophical direction.

Question 2: The status of Conjectures IV.1 and IV.2

Should Conjectures IV.1 and IV.2 turn out to be false — that is, should VR-Sets prove to be substantively **not** equivalent to any countable model of ZFC or AFA — this would be a substantive result. It would mean that the operational ontology yields its own mathematical structure, irreducible to the classical. Open question: to what then does the ZFC-mode of VR-Sets correspond, if not to a countable model of ZFC?

This question is not technical but substantive: it bears on the extent to which VR-Sets is its "own" theory and the extent to which it is a reformulation of the known.

Question 3: Applicability to physics

VR-Numbers (Reznik 2026) discussed possible connections with physics — through $\mathbb{C}$ as the natural algebra of quantum mechanics. VR-Sets potentially deepens this connection: a countable operational ontology may be a suitable frame for discussing quantum states as operational descriptions of observables.

Open direction: is the VR-Sets ontology compatible with the structure of Hilbert space used in quantum mechanics? Hilbert space is classically uncountable; computable Hilbert space (as in computable analysis) is countable. If physics in reality works only with computable states, VR-Sets is a natural ontological frame for quantum mechanics.

This is a speculative direction not developed in the preprint; it is mentioned here as one of the promising areas of application.

Question 4: Relation to the VR programme as a whole

The cycle of works — VR. A Formal System, VR-Numbers, VR-Sets — is built as an ascending sequence: from arithmetic through numerical extensions to set theory. Open question: what is the *next* work in the cycle?

Possible candidates:

- (a) **VR-Analysis** — operational analysis on $\mathbb{R}_{\text{VR-Sets}}$.
- (b) **VR-Categories** — a categorical reconception of the cycle, with connections to topoi and HoTT.
- (c) **VR-Physics** — an operational reformulation of the foundations of physics.
- (d) **VR-Logic** — an extension of the basic logic of VR (two-valued classical) to intuitionistic, modal, or linear operational logics.

The choice of the next work is a substantive question depending on which direction proves most fruitful.

IX.4. Conclusion

The VR-Sets preprint, consisting of nine parts, has set out an operational theory of sets based on the single ontological primitive $\emptyset$. The principal results:

- (1) **The ontological foundation** (Parts I, II, VIII): a set is an operational functionality; membership is reference, not containment; operationality is derivative action upon $\emptyset$, not a self-standing kind of being.
- (2) **The closure principle** (Part II §II.3): the operational replacement for the axioms of existence. Every describable functionality is a set; paradoxical descriptions do not specify functionalities.
- (3) **The ZF axioms as theorems** (Part III): all nine ZF axioms + AC have been derived through the closure principle. Three key divergences from classical ZF: $\wp(\omega)$ is countable; the replacement schema becomes a single theorem; foundation is a mode-dependent property.

(4) **The two modes** (Part IV): the ZFC-mode (well-founded functionalities) and the ZFA-mode (all describable). The distinction is structural, not temporal. Conjectures of equivalence with the classical theories via a countable model.

(5) **VR numbers in VR-Sets** (Part V): von Neumann ordinals as specific well-founded sets; arithmetic inherited without re-proof.

(6) **Cardinality** (Part VI): countability of the entire operational universe; Cantor's diagonal does not operate; AC is a theorem; absence of Banach–Tarski.

(7) **Truth and self-reference** (Part VII): a cautious reformulation of Tarski's hierarchy through the successor operator t . Not a resolution but an operational interpretation.

The position of VR-Sets in the spectrum of foundations of mathematics. VR-Sets is an operationalist position located between formalism and constructivism, closer to computable analysis and predicativism than to Platonism. It does not claim to replace classical ZFC, but provides an alternative ontological frame for mathematics in which every object may be exhibited operationally.

The principal substantive achievement is the demonstration that ontological minimalism ("only $\emptyset$ is, all else is doing") applied to sets yields a coherent, technically workable, and philosophically transparent theory, encompassing the greater part of substantive mathematics and dissolving classical paradoxes (Russell, Banach–Tarski, partially the Liar) ontologically rather than syntactically.

The open questions gathered in this part indicate directions for further work — technical (formalisation in Lean, proofs of conjectures, precise characterisation of describability), substantive (connection with topoi, HoTT, extensions), and philosophical (the nature of "action," the next work in the VR cycle). Each of these directions is fruitful as an independent programme.

Closing the preprint, we repeat the slogan of the cycle:

"Only $\emptyset$ is — all else is doing."

Applied to sets: $\emptyset$ is ontologically primary; everything else is operational action, describable by finite procedures, unfolding into a countable, transparent, coherent mathematical universe. This universe is VR-Sets.

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